NewAnalysis

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2025-06-06

Intepretation: The PCA plot (Dim1: 35.1%, Dim2: 24.6%) reveals distinct clustering patterns in karyotype profiles across treatment groups. Most groups—including Control, G1, G2, K1–K3, and O1–O2—cluster tightly near the origin, indicating similar karyotypic features. In contrast, the K group separates clearly along Dim1 and Dim2, suggesting significant karyotypic alterations in response to treatment. Overall, Dim1 and Dim2 together explain 59.7% of the total variance in the dataset.

Boxplot showing the copy number by chromosome

Observation: The heatmap displays z-score normalized chromosome copy numbers across clones (rows) and chromosomes 1–22 (columns). Variations in color intensity indicate differences in chromosomal copy number, with red suggesting gains and blue indicating losses. Certain chromosomes, such as V4, V8, and V9, show greater variability across clones, suggesting regions of chromosomal instability.

Interpretation: The scree plot shows that the first three principal components explain the majority of the variance in the dataset, with a clear drop after the third component. This suggests that PC1 to PC3 capture the most meaningful karyotypic variation, while subsequent components contribute minimally and likely represent noise.

Objective: To test whether treatment groups differ significantly in their karyotype profiles, using multivariate statistics.

Interpration: A permutational multivariate analysis of variance (PERMANOVA) using Euclidean distance revealed a statistically significant difference in overall karyotype profiles across the 19 treatment groups (F = 22.72, R^2 = 0.351, p = 0.001). This suggests that treatment condition accounts for ~35% of the variance in chromosome-level copy number. However, a test for homogeneity of dispersion (PERMDISP) indicated significant variance heterogeneity among groups ($p < 0.05$), implying that the observed effect may partly reflect differences in within-group variability. These results highlight both treatment-induced karyotype shifts and potential genomic instability under specific conditions.

Interpretation: The test for homogeneity of multivariate dispersions (PERMDISP) was statistically significant (F = 11.41, p = 0.001), indicating that group variances differ in multivariate space. This suggests that the PERMANOVA result (F = 22.72, p = 0.001) should be interpreted with caution, as it may reflect differences in within-group variability as well as differences in centroid (mean) karyotype profiles. Follow-up pairwise dispersion tests are underway to determine which groups contribute most to this variance.

Interpretation: Pairwise tests for multivariate dispersion revealed that several treatment groups — notably P0, R3, and O2 — exhibited significantly higher or lower within-group variability compared to others (e.g., Control, G1, G2). These findings indicate that the significant PERMANOVA result ($p < 0.001$) may be partially attributable to heterogeneity in dispersion, not just shifts in mean karyotype profiles. Consequently, group differences involving these high-dispersion treatments should be interpreted with caution, as they may reflect increased genomic instability rather than systematic karyotype restructuring.

Interpretation: To ensure that our PERMANOVA results were not confounded by differences in group dispersion, we conducted a pairwise Tukey HSD test following a significant PERMDISP result. We identified a subset of groups with homogeneous multivariate dispersion and re-ran PERMANOVA on this subset. The result remained highly significant (F = 22.72, $R^2 = 0.351$, p = 0.001), confirming that treatment-associated differences in karyotype are robust and reflect true shifts in multivariate profile, rather than just variance heterogeneity.

Intepretation: A constrained analysis of principal coordinates (CAP) was conducted to visualize and confirm group-level differences in karyotype structure. The result was highly significant (F = 22.72, p = 0.001), with treatment groups explaining approximately 35.1% of the total variation. This reinforces the findings from PERMANOVA and provides strong multivariate evidence that treatment groups differ in chromosomal copy number profiles.

Interpretation: The CAP (Canonical Analysis of Principal coordinates) plot displays the distribution of samples along the first two canonical axes (CAP1 and CAP2). Most samples cluster tightly around the origin, indicating moderate variation in the constrained ordination space. A few points spread out along CAP1, suggesting some treatment or group differences. Blue “X” markers likely represent group centroids or means, which show partial separation along CAP1. This suggests there may be subtle but meaningful differences in karyotype composition across treatment groups.